COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212	**SECTION:**	CPE31S2
DATE PERFORMED:	07/30/26	**DATE SUBMITTED:**	07/30/26
NAME:	San Juan, Ivan Wayne	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. 1.1 Creating Virtual Machines

1. PROCEDURE OUTPUT

Task 1:

Local Machine

```
sanjuan@sanjuan-VirtualBox:~$ cat /etc/hosts
127.0.0.1       localhost
127.0.0.1       sanjuan-VirtualBox

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
sanjuan@sanjuan-VirtualBox:~$ cat /etc/hostname
Local Machine
```

Server 1

```
sanjuan-server1@sanjuanserver1-VirtualBox:~$ cat /etc/hosts
127.0.0.1       localhost
127.0.0.1       sanjuanserver1-VirtualBox

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
sanjuan-server1@sanjuanserver1-VirtualBox:~$ cat /etc/hostname
Server 1
```

Server 2

```
sanjuan-server2@sanjuanserver2-VirtualBox:~$ cat /etc/hosts
127.0.0.1       localhost
127.0.0.1       sanjuanserver2-VirtualBox

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
sanjuan-server2@sanjuanserver2-VirtualBox:~$ cat /etc/hostname
Server 2
```

Task 2:

Local Machine

```
sanjuan@sanjuan-VirtualBox:~$ sudo apt install openssh-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugin5 libgcab-1.0-0 libsmbios-c2 linux-headers-5.19.0-32-generic linux-hwe-5.19-headers-5.19.0-32
  linux-image-5.19.0-32-generic linux-modules-5.19.0-32-generic linux-modules-extra-5.19.0-32-generic
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  ncurses-term openssh-sftp-server ssh-import-id
Suggested packages:
  molly-guard monkeysphere ssh-askpass
The following NEW packages will be installed:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
0 upgraded, 4 newly installed, 0 to remove and 0 not upgraded.
Need to get 752 kB of archives.
After this operation, 6,051 kB of additional disk space will be used.
Get:1 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 openssh-sftp-server amd64 1:8.9p1-3ubuntu0.16 [38.9 kB]
Get:2 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 openssh-server amd64 1:8.9p1-3ubuntu0.16 [436 kB]
Get:3 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 ncurses-term all 6.3-2ubuntu0.2 [267 kB]
Get:4 http://ph.archive.ubuntu.com/ubuntu jammy/main amd64 ssh-import-id all 5.11-0ubuntu1 [10.1 kB]
Fetched 752 kB in 0s (1,631 kB/s)
Preconfiguring packages ...
Selecting previously unselected package openssh-sftp-server.
(Reading database ... 240805 files and directories currently installed.)
Preparing to unpack .../openssh-sftp-server_1%3a8.9p1-3ubuntu0.16_amd64.deb ...
Unpacking openssh-sftp-server (1:8.9p1-3ubuntu0.16) ...
Selecting previously unselected package openssh-server.
```

```
sanjuan@sanjuan-VirtualBox:~$ sudo service start ssh
start: unrecognized service
sanjuan@sanjuan-VirtualBox:~$ sudo service ssh start
sanjuan@sanjuan-VirtualBox:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
     Active: active (running) since Thu 2026-07-30 09:08:22 PST; 37s ago
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 63802 (sshd)
      Tasks: 1 (limit: 4641)
     Memory: 1.8M
        CPU: 18ms
     CGroup: /system.slice/ssh.service
             └─63802 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 30 09:08:22 sanjuan-VirtualBox systemd[1]: Starting OpenBSD Secure Shell server...
Jul 30 09:08:22 sanjuan-VirtualBox sshd[63802]: Server listening on 0.0.0.0 port 22.
Jul 30 09:08:22 sanjuan-VirtualBox sshd[63802]: Server listening on :: port 22.
Jul 30 09:08:22 sanjuan-VirtualBox systemd[1]: Started OpenBSD Secure Shell server.
sanjuan@sanjuan-VirtualBox:~$ sudo ufw enable
Firewall is active and enabled on system startup
sanjuan@sanjuan-VirtualBox:~$ sudo ufw allow ssh
Rule added
Rule added (v6)
sanjuan@sanjuan-VirtualBox:~$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)
```

Server 1

```
sanjuan-server1@sanjuanserver1-VirtualBox:~$ sudo apt install openssh-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugin5 libgcab-1.0-0 libsmbios-c2 linux-headers-5.19.0-32-generic linux-hwe-5.19-headers-5.19.0-32
  linux-image-5.19.0-32-generic linux-modules-5.19.0-32-generic linux-modules-extra-5.19.0-32-generic
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  ncurses-term openssh-sftp-server ssh-import-id
Suggested packages:
  molly-guard monkeysphere ssh-askpass
The following NEW packages will be installed:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
0 upgraded, 4 newly installed, 0 to remove and 0 not upgraded.
Need to get 752 kB of archives.
After this operation, 6,051 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 openssh-sftp-server amd64 1:8.9p1-3ubuntu0.16 [38.9 kB]
Get:2 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 openssh-server amd64 1:8.9p1-3ubuntu0.16 [436 kB]
Get:3 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 ncurses-term all 6.3-2ubuntu0.2 [267 kB]
Get:4 http://ph.archive.ubuntu.com/ubuntu jammy/main amd64 ssh-import-id all 5.11-0ubuntu1 [10.1 kB]
Fetched 752 kB in 1s (519 kB/s)
Preconfiguring packages ...
Selecting previously unselected package openssh-sftp-server.
(Reading database ... 239530 files and directories currently installed.)
Preparing to unpack .../openssh-sftp-server_1%3a8.9p1-3ubuntu0.16_amd64.deb ...
Unpacking openssh-sftp-server (1:8.9p1-3ubuntu0.16) ...
Selecting previously unselected package openssh-server.
Preparing to unpack .../openssh-server_1%3a8.9p1-3ubuntu0.16_amd64.deb ...
```

```
sanjuan-server1@sanjuanserver1-VirtualBox:~$ sudo service ssh start
sanjuan-server1@sanjuanserver1-VirtualBox:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
     Active: active (running) since Thu 2026-07-30 09:03:39 PST; 2min 9s ago
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 63473 (sshd)
      Tasks: 1 (limit: 4605)
     Memory: 1.7M
        CPU: 17ms
     CGroup: /system.slice/ssh.service
             └─63473 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 30 09:03:39 sanjuanserver1-VirtualBox systemd[1]: Starting OpenBSD Secure Shell server...
Jul 30 09:03:39 sanjuanserver1-VirtualBox sshd[63473]: Server listening on 0.0.0.0 port 22.
Jul 30 09:03:39 sanjuanserver1-VirtualBox sshd[63473]: Server listening on :: port 22.
Jul 30 09:03:39 sanjuanserver1-VirtualBox systemd[1]: Started OpenBSD Secure Shell server.
sanjuan-server1@sanjuanserver1-VirtualBox:~$
```

```
sanjuan-server1@sanjuanserver1-VirtualBox:~$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
sanjuan-server1@sanjuanserver1-VirtualBox:~$ sudo ufw enable
Firewall is active and enabled on system startup
sanjuan-server1@sanjuanserver1-VirtualBox:~$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)

sanjuan-server1@sanjuanserver1-VirtualBox:~$
```

Server 2

```
sanjuan-server2@sanjuanserver2-VirtualBox:~$ sudo apt install openssh-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugin5 libgcab-1.0-0 libsmbios-c2 linux-headers-5.19.0-32-generic linux-hwe-5.19-headers-5.19.0-32
  linux-image-5.19.0-32-generic linux-modules-5.19.0-32-generic linux-modules-extra-5.19.0-32-generic
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  ncurses-term openssh-sftp-server ssh-import-id
Suggested packages:
  molly-guard monkeysphere ssh-askpass
The following NEW packages will be installed:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
0 upgraded, 4 newly installed, 0 to remove and 0 not upgraded.
Need to get 752 kB of archives.
After this operation, 6,051 kB of additional disk space will be used.
Get:1 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 openssh-sftp-server amd64 1:8.9p1-3ubuntu0.16 [38.9 kB]
Get:2 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 openssh-server amd64 1:8.9p1-3ubuntu0.16 [436 kB]
Get:3 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 ncurses-term all 6.3-2ubuntu0.2 [267 kB]
Get:4 http://ph.archive.ubuntu.com/ubuntu jammy/main amd64 ssh-import-id all 5.11-0ubuntu1 [10.1 kB]
Fetched 752 kB in 0s (1,909 kB/s)
Preconfiguring packages ...
Selecting previously unselected package openssh-sftp-server.
(Reading database ... 239341 files and directories currently installed.)
Preparing to unpack .../openssh-sftp-server_1%3a8.9p1-3ubuntu0.16_amd64.deb ...
Unpacking openssh-sftp-server (1:8.9p1-3ubuntu0.16) ...
Selecting previously unselected package openssh-server.
Preparing to unpack .../openssh-server_1%3a8.9p1-3ubuntu0.16_amd64.deb ...
Unpacking openssh-server (1:8.9p1-3ubuntu0.16) ...
Selecting previously unselected package ncurses-term.
Preparing to unpack .../ncurses-term_6.3-2ubuntu0.2_all.deb ...
Unpacking ncurses-term (6.3-2ubuntu0.2) ...
Selecting previously unselected package ssh-import-id.
Preparing to unpack .../ssh-import-id_5.11-0ubuntu1_all.deb ...
Unpacking ssh-import-id (5.11-0ubuntu1) ...
```

```
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
sanjuan-server2@sanjuanserver2-VirtualBox:~$ sudo service ssh start
sanjuan-server2@sanjuanserver2-VirtualBox:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
     Active: active (running) since Thu 2026-07-30 09:08:35 PST; 2min 56s ago
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 61804 (sshd)
      Tasks: 1 (limit: 4641)
     Memory: 1.7M
        CPU: 20ms
     CGroup: /system.slice/ssh.service
             └─61804 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 30 09:08:35 sanjuanserver2-VirtualBox systemd[1]: Starting OpenBSD Secure Shell server...
Jul 30 09:08:35 sanjuanserver2-VirtualBox sshd[61804]: Server listening on 0.0.0.0 port 22.
Jul 30 09:08:35 sanjuanserver2-VirtualBox sshd[61804]: Server listening on :: port 22.
Jul 30 09:08:35 sanjuanserver2-VirtualBox systemd[1]: Started OpenBSD Secure Shell server.
sanjuan-server2@sanjuanserver2-VirtualBox:~$ sudo ufw enable
Firewall is active and enabled on system startup
sanjuan-server2@sanjuanserver2-VirtualBox:~$ sudo ufw allow ssh
Rule added
Rule added (v6)
sanjuan-server2@sanjuanserver2-VirtualBox:~$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)
```

Task 3:
HOST -> Server 1 and 2

```
sanjuan@sanjuan-VirtualBox:~$ ping 192.168.56.118
PING 192.168.56.118 (192.168.56.118) 56(84) bytes of data.
64 bytes from 192.168.56.118: icmp_seq=1 ttl=64 time=2.09 ms
64 bytes from 192.168.56.118: icmp_seq=2 ttl=64 time=0.713 ms
64 bytes from 192.168.56.118: icmp_seq=3 ttl=64 time=4.89 ms
64 bytes from 192.168.56.118: icmp_seq=4 ttl=64 time=0.499 ms
64 bytes from 192.168.56.118: icmp_seq=5 ttl=64 time=0.460 ms
^C
--- 192.168.56.118 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 9103ms
rtt min/avg/max/mdev = 0.460/1.729/4.889/1.689 ms
sanjuan@sanjuan-VirtualBox:~$ ping 192.168.56.119
PING 192.168.56.119 (192.168.56.119) 56(84) bytes of data.
64 bytes from 192.168.56.119: icmp_seq=1 ttl=64 time=1.81 ms
64 bytes from 192.168.56.119: icmp_seq=2 ttl=64 time=0.422 ms
64 bytes from 192.168.56.119: icmp_seq=3 ttl=64 time=4.27 ms
^C
--- 192.168.56.119 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 3963ms
rtt min/avg/max/mdev = 0.422/2.168/4.274/1.592 ms
```

Server 1 -> Server 2

```
sanjuan-server1@sanjuanserver1-VirtualBox:~$ ping 192.168.56.118
PING 192.168.56.118 (192.168.56.118) 56(84) bytes of data.
64 bytes from 192.168.56.118: icmp_seq=1 ttl=64 time=2.25 ms
64 bytes from 192.168.56.118: icmp_seq=2 ttl=64 time=1.04 ms
64 bytes from 192.168.56.118: icmp_seq=3 ttl=64 time=0.614 ms
^C
--- 192.168.56.118 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 7269ms
rtt min/avg/max/mdev = 0.614/1.300/2.252/0.694 ms
```

Task 4:
Server 1

```
sanjuan@sanjuan-VirtualBox:~$ ssh sanjuan-server1@192.168.56.119
The authenticity of host '192.168.56.119 (192.168.56.119)' can't be established.
ED25519 key fingerprint is SHA256:cGdQwX6FL5UAGTHW3444vpkjKZh68GxwEIc5XSecdeU.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.119' (ED25519) to the list of known hosts.
sanjuan-server1@192.168.56.119's password:
Permission denied, please try again.
sanjuan-server1@192.168.56.119's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-134-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

10 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

*** System restart required ***

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
```

```
sanjuan@sanjuan-VirtualBox:~$ ssh sanjuan-server1@server1
sanjuan-server1@server1's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-134-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

10 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

*** System restart required ***
Last login: Thu Jul 30 09:26:52 2026 from 192.168.56.117
sanjuan-server1@sanjuanserver1-VirtualBox:~$
logout
Connection to server1 closed
```

Server 2:

```
sanjuan@sanjuan-VirtualBox:~$ ssh sanjuan-server2@192.168.56.118
The authenticity of host '192.168.56.118 (192.168.56.118)' can't be established.
ED25519 key fingerprint is SHA256:LHQ5gRqK8DoR9cTB/9TJ/2/1jgVq52ds4DtiqC+RZLI.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.118' (ED25519) to the list of known hosts.
sanjuan-server2@192.168.56.118's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-134-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

10 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

*** System restart required ***

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

sanjuan-server2@sanjuanserver2-VirtualBox:~$
logout
Connection to server2 closed.
sanjuan@sanjuan-VirtualBox:~$ ssh sanjuan-server2@server2
The authenticity of host 'server2 (192.168.56.118)' can't be established.
ED25519 key fingerprint is SHA256:LHQ5gRqK8DoR9cTB/9TJ/2/1jgVq52ds4DtiqC+RZLI.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:4: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'server2' (ED25519) to the list of known hosts.
sanjuan-server2@server2's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-134-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

10 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

*** System restart required ***
Last login: Thu Jul 30 09:28:36 2026 from 192.168.56.117
sanjuan-server2@sanjuanserver2-VirtualBox:~$
```

COMPUTER ENGINEERING

QUEZON CITY

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

1. How are we able to use the hostname instead of IP address in SSH commands?
- We can use the hostname instead of the IP address because the computer uses the DNS (Domain Name System) to find the correct IP address automatically.

2. How secured is SSH?
- SSH is very secure because it encrypts the data sent between computers, making it difficult for others to read or steal the information.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **ChatGPT**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Grammar correction**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I use it to generate a much better grammar**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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